

DevOps Portfolio

End-to-End Documentation — From Scratch

Terraform AWS Infra • Ansible Config Mgmt • CI/CD Docker & Kubernetes

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GitHub	https://github.com/sravanigaddam-devops Repos: terraform-aws-infra, ansible-config-mgmt, cicd-python-app-k8s
Docker Hub	sravani81299/sravani-app — tags: 1.0.0, 1.0.1, latest (sha256:18d0...)
Live Infra (before destroy)	VPC vpc-03b31ff28e41bc125 EC2 i-00f507eb0af08bb0e 3.231.160.92 t3.micro kind v1.37.0 3/3 Running
Stack	AWS us-east-1, Terraform 1.15.8, Ansible 2.21.3, Python 3.13 Flask 3.1, Docker 29.7.2, kind 0.33.0, Jenkins
Date	2026-08-29 — All 3 projects executed live, screenshots & outputs captured

This document explains **every file** from scratch (app.py, requirements.txt, test_app.py, Jenkinsfile, deployment.yaml, service.yaml, Dockerfile, inventory, playbooks, roles, ansible.cfg, envs/dev, terraform.tfvars, modules, backend.tf, main.tf, outputs.tf, provider.tf, variables.tf) and **every command** used — copy-paste reproducible. Proof screenshots and live outputs are embedded.

DevOps Portfolio — 3 Projects Architecture | sravani-devops | us-east-1
Terraform AWS Infra • Ansible Config Mgmt • CI/CD Docker & K8s (sravani81299)

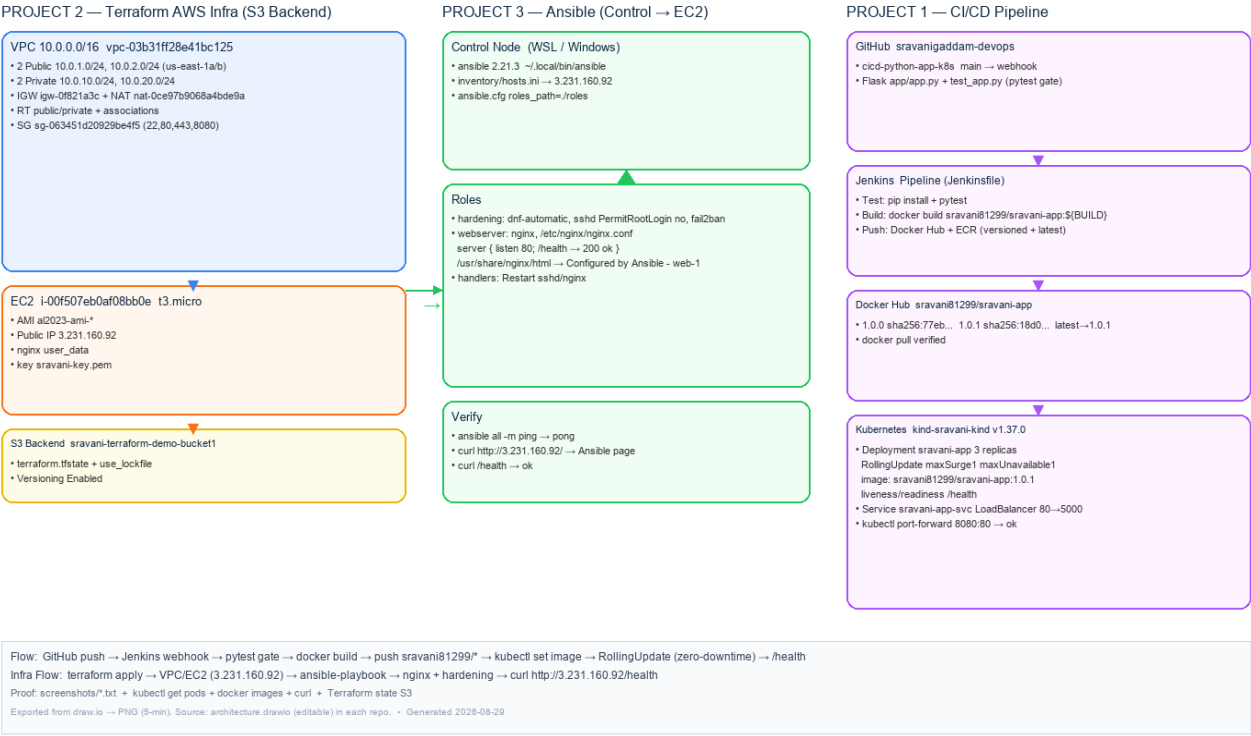


Figure 1 — 5-min architecture (draw.io → PNG) — editable as architecture.drawio in each repo. Generated via generate_diagram.py:1

Table of Contents

1. At-a-Glance & Execution Order	3
2. Architecture Deep Dive	4
3. Project 2 — Terraform AWS Infra (Every File)	5
4. Project 3 — Ansible Config Mgmt (Every File)	9
5. Project 1 — CI/CD Docker & K8s (Every File)	12
6. Commands Executed — Copy-Paste Log	16
7. Live Outputs & Proof	19
8. Troubleshooting — 7 Real Errors Fixed	21
9. Interview Guide — 30s / 2 / 5 / 10 min + Q&A;	22
10. Repos, Cleanup & Next Steps	24

How to use: Follow §3→4→5 in order. Each file has *Purpose* → *Content* → *Interview line*. Commands in §6 are exactly as run on 2026-08-29 (PowerShell + WSL). Screenshots in each repo *screenshots/*.txt* are proof.

1. At-a-Glance & Execution Order

Order **2** → **3** → **1** is intentional: Terraform creates the network, Ansible configures the hosts, CI/CD deploys the app onto that infra. In interview, explain in this order.

#	Project	What It Proves	Key Files	Live Output
2	Terraform AWS Infra	IaC, modules, S3 remote state + lock, safe plan/apply	provider.tf, backend.tf, variables.tf, main.tf, outputs.tf, modules/vpc sg ec2	vpc-03b31ff28e41bc125, 3.231.160.92
3	Ansible Config Mgmt	Idempotent, roles, 1→N, hardening	ansible.cfg, inventory/hosts.ini, playbooks/site.yml, roles/hardening, roles/webserver	http://3.231.160.92 Configured by Ansible
1	CI/CD Docker & K8s	Gated, versioned, RollingUpdate	app.py, requirements.txt, test_app.py, Dockerfile, k8s/deployment.yaml, service.yaml, Jenkinsfile	sravani81299:1.0.1 3/3 Running

2. Architecture Deep Dive

See Figure 1. **Infra path:** terraform apply → VPC 10.0.0.0/16 (2 public 10.0.1/2.0/24 + 2 private 10.0.10/20.0/24, us-east-1a/b) → IGW igw-0f821a3c + NAT nat-0ce97b9068a4bde9a + RTs → SG sg-063451d20929be4f5 (22,80,443,8080) → EC2 t3.micro 3.231.160.92 nginx user_data → S3 sravani-terraform-demo-bucket1/terraform.tfstate use_lockfile

Config path: ansible-playbook playbooks/site.yml -i inventory/hosts.ini (ANSIBLE_ROLES_PATH=./roles) from WSL → 3.231.160.92 → hardening (dnf-automatic, sshd PermitRootLogin no) + webserver (nginx.conf server {listen 80; /health 200 ok}, index.html)

App path: GitHub push sravanigaddam-devops/cicd-python-app-k8s main → Jenkins webhook → pytest → docker build sravani81299/sravani-app:\${BUILD_NUMBER}+latest → kind load → kubectl set image → RollingUpdate maxSurge1 maxUnavailable1 → Service LoadBalancer 80→5000 → port-forward 8080:80 → curl /health

DevOps Portfolio — 3 Projects Architecture | sravani-devops | us-east-1

Terraform AWS Infra • Ansible Config Mgmt • CI/CD Docker & K8s (sravani81299)

PROJECT 2 — Terraform AWS Infra (S3 Backend)

VPC 10.0.0.0/16 vpc-03b31f28e41bc125

- 2 Public 10.0.1.0/24, 10.0.2.0/24 (us-east-1a/b)
- 2 Private 10.0.10.0/24, 10.0.20.0/24
- IGW igw-0f821a3c + NAT nat-0ce97b9068a4bde9a
- RT public/private + associations
- SG sg-063451d20529be4f5 (22,80,443,8080)

EC2 i-00f507eb0af08bb0e t3.micro

- AMI a2023-ami*
- Public IP 3.231.160.92
- nginx user_data
- key sravani-key.pem

S3 Backend sravani-terraform-demo-bucket1

- terraform.tfstate + use_lockfile
- Versioning Enabled

PROJECT 3 — Ansible (Control → EC2)

Control Node (WSL / Windows)

- ansible 2.21.3 ~/local/bin/ansible
- inventory/hosts.ini → 3.231.160.92
- ansible.cfg roles_path=~/roles

Roles

- hardening: dnf-automatic, sshd PermitRootLogin no, fail2ban
- webserver: nginx, let's/nginx/nginx.conf
- server { listen 80; /health → 200 ok }
- /usr/share/nginx/html → Configured by Ansible - web-1
- handlers: Restart sshd/nginx

Verify

- ansible all -m ping → pong
- curl http://3.231.160.92/ → Ansible page
- curl /health → ok

PROJECT 1 — CI/CD Pipeline

GitHub sravanigaddam-devops

- cicd-python-app-k8s main → webhook
- Flask app/app.py + test_app.py (pytest gate)

Jenkins Pipeline (Jenkinsfile)

- Test: pip install + pytest
- Build: docker build sravani81299/sravani-app:\$[BUILD]
- Push: Docker Hub + ECR (versioned + latest)

Docker Hub sravani81299/sravani-app

- 1.0.0 sha256:77eb... 1.0.1 sha256:18d0... latest → 1.0.1
- docker pull verified

Kubernetes kind-sravani-kind v1.37.0

- Deployment sravani-app 3 replicas
- RollingUpdate maxSurge1 maxUnavailable1
- image: sravani81299/sravani-app:1.0.1
- liveness/readiness /health
- Service sravani-app-svc LoadBalancer 80 → 5000
- kubectl port-forward 8080:80 → ok

Flow: GitHub push → Jenkins webhook → pytest gate → docker build → push sravani81299 → kubectl set image → RollingUpdate (zero-downtime) → /health

Infra Flow: terraform apply → VPC/EC2 (3.231.160.92) → ansible-playbook → nginx + hardening → curl http://3.231.160.92/health

Proof: screenshots/*.txt • kubectl get pods + docker images + curl + Terraform state S3

Exported from draw.io → PNG (5-min). Source: architecture.drawio (editable) in each repo. • Generated 2026-08-29

Figure 1 repeated — also in each repo as architecture.png + architecture.drawio

3. Project 2 — Terraform AWS Infra (Every File)

Goal: Repeatable AWS network + EC2 with modules and S3 remote state. No click-ops. *terraform fmt* → *init* → *validate* → *plan* → *out* → *apply* is the safe gate.

Directory Tree

```

terraform-aws-infra/
  provider.tf          # AWS provider ~5.0 + default_tags
  backend.tf           # S3 use_lockfile
  variables.tf         # all inputs
  terraform.tfvars.example # example values
  terraform.tfvars     # actual (key_name=sravani-key, t3.micro)
  main.tf              # composition root
  outputs.tf           # outputs
  modules/vpc/         # VPC, subnets, IGW, NAT, RT
  modules/security-group/ # SG dynamic ingress
  modules/ec2/         # EC2 + user_data
  envs/dev/terraform.tfvars # dev overlay
  .gitignore, README.md, INTERVIEW.md, architecture.png/drawio, screenshots/

```

provider.tf

Declares `required_version >=1.5.0` and `aws ~>5.0`. Provider `aws` `region = var.aws_region`. `default_tags` adds `Project/Env/ManagedBy` to every resource for cost tracking. Interview: 'I pin provider major to avoid breaking changes; `default_tags` enforces tagging.'

terraform-aws-infra/provider.tf

```

terraform {
  required_version = ">= 1.5.0"
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = var.aws_region
  default_tags {

```

Proof: screenshots/*.txt • Architecture: architecture.png/drawio • GitHub: sravanigaddam-devops • Docker Hub: sravani81299

```

tags = {
  Project      = var.project_name
  Environment  = var.environment
  ManagedBy    = "terraform"
}
}
}

```

backend.tf

S3 backend bucket=sravani-terraform-demo-bucket1 key=terraform-aws-infra/terraform.tfstate region=us-east-1 encrypt=true use_lockfile=true (S3 native locking, replaces DynamoDB, TF >=1.10). Commented local backend for testing. Interview: 'I hit 412 PreconditionFailed on .tflock — versioned bucket left delete markers; deleted all versions + force-unlock.'

terraform-aws-infra/backend.tf

```

# Remote state - S3 backend with native S3 locking (Terraform >=1.10)
# Bucket must exist beforehand. Create it with:
# aws s3api create-bucket --bucket sravani-terraform-demo-bucket1 --region us-east-1
# aws s3api put-bucket-versioning --bucket sravani-terraform-demo-bucket1 --versioning-
configuration Status=Enabled
# aws s3api put-bucket-encryption --bucket sravani-terraform-demo-bucket1 --server-side-
encryption-configuration '{"Rules":[{"ApplyServerSideEncryptionByDefault":{"SSEAlgorithm":
"AES256"}}]}'

terraform {
  backend "s3" {
    bucket      = "sravani-terraform-demo-bucket1"
    key         = "terraform-aws-infra/terraform.tfstate"
    region      = "us-east-1"
    encrypt     = true
    use_lockfile = true # S3 native locking, replaces DynamoDB
  }
}

# Alternative: local backend for initial testing (comment out s3 backend above and uncomment
below)
# terraform {
#   backend "local" {
#     path = "terraform.tfstate"
#   }
# }

```

variables.tf

Inputs: aws_region us-east-1, project_name sravani-devops, environment dev, vpc_cidr 10.0.0.0/16, public/private_subnet_cidrs 2 each, AZs us-east-1a/b, instance_type t3.micro (was t2.micro → not free-tier, switched after describe-instance-types), key_name (null→sravani-key), allowed_ssh_cidr 0.0.0.0/0 (tighten to MY_IP/32). Interview: 'Variables make it env-agnostic; I overrode via tfvars and -var-file=envs/dev.'

terraform-aws-infra/variables.tf

```

variable "aws_region" {
  description = "AWS region"
  type        = string
  default     = "us-east-1"
}

variable "project_name" {
  description = "Project name prefix"
  type        = string
  default     = "sravani-devops"
}

variable "environment" {
  description = "Environment (dev/prod)"
  type        = string
  default     = "dev"
}

variable "vpc_cidr" {
  description = "VPC CIDR"
  type        = string
  default     = "10.0.0.0/16"
}

variable "public_subnet_cidrs" {
  type        = list(string)
  default     = ["10.0.1.0/24", "10.0.2.0/24"]
}

```

```

variable "private_subnet_cidrs" {
  type    = list(string)
  default = ["10.0.10.0/24", "10.0.20.0/24"]
}

variable "availability_zones" {
  type    = list(string)
  default = ["us-east-1a", "us-east-1b"]
}

variable "instance_type" {
  type    = string
  default = "t3.micro"
}

variable "key_name" {
  description = "EC2 key pair name (optional)"
  type        = string
  default     = null
}

... (5 more lines)

```

terraform.tfvars.example

Documented example values. Copy to terraform.tfvars and set key_name to your EC2 key pair. Shows plan/apply workflow.

terraform-aws-infra\terraform.tfvars.example

```

aws_region    = "us-east-1"
project_name  = "sravani-devops"
environment   = "dev"

vpc_cidr      = "10.0.0.0/16"
public_subnet_cidrs = ["10.0.1.0/24", "10.0.2.0/24"]
private_subnet_cidrs = ["10.0.10.0/24", "10.0.20.0/24"]
availability_zones = ["us-east-1a", "us-east-1b"]

instance_type = "t2.micro"
key_name      = null          # set to your EC2 key pair name if you have one
allowed_ssh_cidr = "0.0.0.0/0" # change to YOUR_IP/32 for security

```

terraform.tfvars

Actual values used: aws_region us-east-1, project_name sravani-devops, env dev, t3.micro, key_name sravani-key. Created manually to enable Ansible SSH.

terraform-aws-infra\terraform.tfvars

```

aws_region    = "us-east-1"
project_name  = "sravani-devops"
environment   = "dev"
instance_type = "t3.micro"
key_name      = "sravani-key"
allowed_ssh_cidr = "0.0.0.0/0"

```

main.tf

Composition: module vpc (cidr, subnets, AZs), module web_sg (dynamic ingress 22,80,443,8080), data aws_ami amazon owner amazon al2023-ami-*x86_64, module web_ec2 (subnet public[0], SG, key_name, user_data base64 nginx). Interview: 'Modules make VPC reusable; dynamic ingress lets me add ports without copy-paste; data source always picks latest AL2023.'

terraform-aws-infra\main.tf

```

module "vpc" {
  source = "../modules/vpc"

  project_name = var.project_name
  environment  = var.environment
  vpc_cidr     = var.vpc_cidr
  public_subnet_cidrs = var.public_subnet_cidrs
  private_subnet_cidrs = var.private_subnet_cidrs
  availability_zones = var.availability_zones
}

module "web_sg" {
  source = "../modules/security-group"

  name          = "${var.project_name}-${var.environment}-web-sg"
  description    = "Security group for web servers"
  vpc_id        = module.vpc.vpc_id

```

```

ingress_rules = [
  {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = [var.allowed_ssh_cidr]
    description = "SSH"
  },
  {
    from_port = 80
    to_port   = 80
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
    description = "HTTP"
  },
  {
    from_port = 443
    to_port   = 443
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
    description = "HTTPS"
  },
  {
    from_port = 8080
    to_port   = 8080
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
    description = "App port"
  }
]

environment = var.environment
... (41 more lines)

```

outputs.tf

Exports `vpc_id`, `public_subnet_ids`, `private_subnet_ids`, `web_sg_id`, `ec2_public_ip/dns/instance_id` for Ansible inventory. Used as: `terraform output -json | jq .ec2_public_ip.value`

terraform-aws-infra\outputs.tf

```

output "vpc_id" {
  value = module.vpc.vpc_id
}

output "public_subnet_ids" {
  value = module.vpc.public_subnet_ids
}

output "private_subnet_ids" {
  value = module.vpc.private_subnet_ids
}

output "web_sg_id" {
  value = module.web_sg.security_group_id
}

output "ec2_public_ip" {
  value = module.web_ec2.public_ip
}

output "ec2_public_dns" {
  value = module.web_ec2.public_dns
}

output "ec2_instance_id" {
  value = module.web_ec2.instance_id
}

```

provider.tf (envs/dev)

Overlay per env. Can run: `terraform plan -var-file=envs/dev/terraform.tfvars` to separate dev/prod.

terraform-aws-infra\envs\dev\terraform.tfvars

```

aws_region    = "us-east-1"
project_name  = "sravani-devops"
environment   = "dev"
instance_type = "t2.micro"

```

modules/vpc/main.tf

Creates `aws_vpc` (`enable_dns_hostnames/support`), `aws_internet_gateway`, `aws_subnet` `public[2]` `map_public_ip_on_launch=true`, `private[2]`, `aws_eip` `nat[1]`, `aws_nat_gateway`, `aws_route_table` `public` (`0.0.0.0/0`→IGW) + `private` (→NAT), associations. `Count = len(subnet_cidrs)`. Interview: 'Count makes it scale to 3 AZs; NAT in public subnet gives private egress.'

terraform-aws-infra/modules/vpc/main.tf

```
resource "aws_vpc" "main" {
  cidr_block      = var.vpc_cidr
  enable_dns_hostnames = true
  enable_dns_support = true

  tags = {
    Name = "${var.project_name}-${var.environment}-vpc"
  }
}

resource "aws_internet_gateway" "main" {
  vpc_id = aws_vpc.main.id
  tags = { Name = "${var.project_name}-${var.environment}-igw" }
}

resource "aws_subnet" "public" {
  count          = length(var.public_subnet_cidrs)
  vpc_id        = aws_vpc.main.id
  cidr_block     = var.public_subnet_cidrs[count.index]
  availability_zone = var.availability_zones[count.index]
  map_public_ip_on_launch = true

  tags = { Name = "${var.project_name}-${var.environment}-public-${count.index + 1}" }
}

resource "aws_subnet" "private" {
  count          = length(var.private_subnet_cidrs)
  vpc_id        = aws_vpc.main.id
  cidr_block     = var.private_subnet_cidrs[count.index]
  availability_zone = var.availability_zones[count.index]

  tags = { Name = "${var.project_name}-${var.environment}-private-${count.index + 1}" }
}

resource "aws_eip" "nat" {
  count = 1
  domain = "vpc"
  tags = { Name = "${var.project_name}-${var.environment}-nat-eip" }
}

resource "aws_nat_gateway" "main" {
  allocation_id = aws_eip.nat[0].id
  subnet_id     = aws_subnet.public[0].id
  tags          = { Name = "${var.project_name}-${var.environment}-nat" }
  depends_on    = [aws_internet_gateway.main]
}

resource "aws_route_table" "public" {
  vpc_id = aws_vpc.main.id
  route {
    ... (26 more lines)
  }
}
```

modules/vpc/variables.tf & outputs.tf

Variables: `project_name`, `env`, `vpc_cidr`, `public/private_subnet_cidrs`, AZs. Outputs: `vpc_id`, `public/private_subnet_ids`, `nat_gateway_id`.

terraform-aws-infra/modules/vpc/variables.tf

```
variable "project_name" { type = string }
variable "environment" { type = string }
variable "vpc_cidr" { type = string }
variable "public_subnet_cidrs" { type = list(string) }
variable "private_subnet_cidrs" { type = list(string) }
variable "availability_zones" { type = list(string) }
```

modules/security-group/main.tf

Dynamic ingress block over `var.ingress_rules` (`from_port/to_port/protocol/cidr_blocks/description`). Egress allow all. Tags `Name/Env`. Interview: 'Dynamic blocks avoid repeating ingress stanzas.'

terraform-aws-infra/modules/security-group/main.tf

```
resource "aws_security_group" "main" {
```



```

name      = var.name
description = var.description
vpc_id     = var.vpc_id

dynamic "ingress" {
  for_each = var.ingress_rules
  content {
    description = lookup(ingress.value, "description", null)
    from_port   = ingress.value.from_port
    to_port     = ingress.value.to_port
    protocol    = ingress.value.protocol
    cidr_blocks = lookup(ingress.value, "cidr_blocks", null)
  }
}

egress {
  from_port = 0
  to_port   = 0
  protocol  = "-1"
  cidr_blocks = ["0.0.0.0/0"]
}

tags = { Name = var.name, Environment = var.environment }
}

```

modules/ec2/main.tf

aws_instance with ami, instance_type, subnet_id, vpc_security_group_ids, key_name, user_data
base64decode(var.user_data). Tags merge Env. Interview: 'User_data installs nginx; I base64 encode in root then decode in module to keep plain text readable.'

terraform-aws-infra/modules/ec2/main.tf

```

resource "aws_instance" "main" {
  ami           = var.ami_id
  instance_type = var.instance_type
  subnet_id     = var.subnet_id
  vpc_security_group_ids = var.security_groups
  key_name      = var.key_name
  user_data     = var.user_data != null ? base64decode(var.user_data) : null

  tags = merge({ Environment = var.environment }, var.tags)
}

```

.gitignore

Python/Node+Terraform: .terraform/, *.tfstate*, tfplan, .terraform.lock.hcl, node_modules/, __pycache__/, .env, *.pem, *.key, !terraform.tfvars.example. Keeps secrets out.

terraform-aws-infra/.gitignore

```

# Terraform
.terraform/
.terraform.lock.hcl
*.tfstate
*.tfstate.*
*.tfplan
tfplan
.terraform.tfstate.lock.info
terraform.tfstate.tflock
crash.log
crash.*.log
override.tf
override.tf.json
*_override.tf
*_override.tf.json
.terraformrc
terraform.rc

# Python (Flask app if used with infra scripts)
__pycache__/
*.py[cod]
*$py.class
*.pyo
.pytest_cache/
.venv/
venv/
env/
ENV/

# Node

```

```

node_modules/
npm-debug.log*
yarn-debug.log*
yarn-error.log*
dist/
build/

# Secrets & env
.env
.env.*
*.pem
*.key
*.crt
*.tfvars
!terraform.tfvars.example
!envs/**/*.*tfvars.example
secrets/
credentials

# OS & IDE
... (7 more lines)

```

4. Project 3 — Ansible Config Mgmt (Every File)

Goal: One control node idempotently configures N EC2s. Roles separate hardening vs app. Verified by *ansible ping* → *pong* and *curl /health*.

Directory Tree

```

ansible-config-mgmt/
  ansible.cfg                # defaults
  inventory/hosts.ini        # web-1 3.231.160.92
  playbooks/site.yml         # pre_tasks + roles + post
  roles/hardening/tasks/main.yml + handlers/main.yml
  roles/webserver/tasks/main.yml + handlers/main.yml
  .gitignore, README.md, architecture.png/drawio, screenshots/

```

ansible.cfg

defaults: inventory=inventory/hosts.ini, host_key_checking=False (for EC2), retry_files_enabled=False, stdout_callback=yaml, roles_path=./roles; [privilege_escalation] become=True sudo. Note world-writable /mnt/c ignores this → needed ANSIBLE_ROLES_PATH=./roles. Interview: 'I set yaml callback for readable diffs; roles_path is why hardening was not found initially.'

ansible-config-mgmt\ansible.cfg

```

[defaults]
inventory = inventory/hosts.ini
host_key_checking = False
retry_files_enabled = False
stdout_callback = yaml
roles_path = ./roles

[privilege_escalation]
become = True
become_method = sudo
become_user = root
become_ask_pass = False

```

inventory/hosts.ini

Group [webservers] web-1 ansible_host=3.231.160.92 ansible_user=ec2-user
 ansible_ssh_private_key_file=~/.ssh/sravani-key.pem (Windows: C:/...pem, WSL: ~/.ssh/...). [webservers:vars]
 ansible_python_interpreter=/usr/bin/python3. Generated from terraform output -json. Add web-2 for scale. Interview: 'Inventory is static here; for auto-scale I'd use dynamic inventory from terraform tags.'

ansible-config-mgmt\inventory\hosts.ini

```

[webservers]
web-1 ansible_host=3.231.160.92 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/
sravani-key.pem

[webservers:vars]
ansible_python_interpreter=/usr/bin/python3

```

playbooks/site.yml

Play Configure web servers: hosts webserver become true, pre_tasks dnf update_cache, roles hardening, webserver, post_tasks ensure nginx started/enabled. Interview: 'Pre/post tasks guarantee cache and service; roles keep playbook thin.'

ansible-config-mgmt\playbooks\site.yml

```
---
- name: Configure web servers
  hosts: webserver
  become: true

  pre_tasks:
    - name: Ensure dnf cache is updated (Amazon Linux 2023)
      dnf:
        update_cache: true
        when: ansible_os_family == "RedHat"

  roles:
    - hardening
    - webserver

  post_tasks:
    - name: Ensure nginx is running
      systemd:
        name: nginx
        state: started
        enabled: true
```

roles/hardening/tasks/main.yml

Tasks: dnf-automatic present, lineinfile sshd_config PermitRootLogin no notify Restart sshd, fail2ban present ignore_errors, systemd disable postfix loop ignore_errors. Interview: 'Handler restarts sshd only if file changed; ignore_errors for optional packages.'

ansible-config-mgmt\roles\hardening\tasks\main.yml

```
---
- name: Ensure automatic security updates are enabled
  dnf:
    name: dnf-automatic
    state: present

- name: Harden sshd - disable root login
  lineinfile:
    path: /etc/ssh/sshd_config
    regexp: '^#?PermitRootLogin'
    line: 'PermitRootLogin no'
    notify: Restart sshd

- name: Ensure fail2ban is installed (if available)
  dnf:
    name: fail2ban
    state: present
  ignore_errors: true

- name: Disable unused services
  systemd:
    name: "{{ item }}"
    enabled: false
    state: stopped
  loop:
    - postfix
  ignore_errors: true
```

roles/hardening/handlers/main.yml

Handler Restart sshd: systemd name=sshd state=restarted. Was missing → added after ERROR handler not found.

ansible-config-mgmt\roles\hardening\handlers\main.yml

```
---
- name: Restart sshd
  systemd:
    name: sshd
    state: restarted
```

roles/webserver/tasks/main.yml

Tasks: dnf nginx present, copy nginx.conf (user nginx; events; http { server listen 80; root /usr/share/nginx/html; /health return 200 ok }) notify Restart nginx, copy index.html Configured by Ansible - {{inventory_hostname}} notify, firewall tasks (http/https/8080). Fixes: copy notify indent (task level not inside copy), firewall open. Interview: 'Notify ensures nginx restarts

only on config change; firewalld caused connection refused until I opened http.'

ansible-config-mgmt\roles\webserver\tasks\main.yml

```
---
- name: Install nginx
  dnf:
    name: nginx
    state: present

- name: Deploy nginx config
  copy:
    dest: /etc/nginx/nginx.conf
    content: |
      user nginx;
      worker_processes auto;
      error_log /var/log/nginx/error.log;
      pid /run/nginx.pid;
      events { worker_connections 1024; }
      http {
        include /etc/nginx/mime.types;
        default_type application/octet-stream;
        log_format main '$remote_addr - $remote_user [$time_local] "$request"';
        access_log /var/log/nginx/access.log main;
        server {
          listen 80;
          root /usr/share/nginx/html;
          index index.html;
          location /health { return 200 'ok'; add_header Content-Type text/plain; }
        }
      }
  notify: Restart nginx

- name: Deploy index.html
  copy:
    dest: /usr/share/nginx/html/index.html
    content: |
      <html><body>
      <h1>Configured by Ansible - {{ inventory_hostname }}</h1>
      <p>Project: sravani-devops | Env: dev</p>
      <p>Date: {{ ansible_date_time.iso8601 }}</p>
      </body></html>
  notify: Restart nginx

- name: Ensure firewalld is running
  systemd:
    name: firewalld
    state: started
    enabled: true
... (20 more lines)
```

roles/webserver/handlers/main.yml

Handler Restart nginx: systemd name=nginx state=restarted.

ansible-config-mgmt\roles\webserver\handlers\main.yml

```
---
- name: Restart nginx
  systemd:
    name: nginx
    state: restarted
```

.gitignore

Ansible *.retry, .vault_pass + Python/Node + secrets *.pem, .env, secrets/ + OS .DS_Store.

ansible-config-mgmt\gitignore

```
# Ansible
*.retry
*.vault_pass
.vault_pass
vault_pass.txt
.hosts
inventory/hosts.ini.bak

# Python
__pycache__/
*.py[cod]
*$py.class
.pytest_cache/
.venv/
```

```

venv/
env/

# Node
node_modules/
npm-debug.log*
yarn-debug.log*

# Secrets & env
.env
.env.*
*.pem
*.key
*.vault
secrets/
credentials/
ansible.log

# Terraform crossover (if inventory generated from terraform)
*.tfstate
*.tfstate.*

# OS & IDE
.DS_Store
Thumbs.db
.vscode/
.idea/
*.swp
*.log

```

5. Project 1 — CI/CD Docker & K8s (Every File)

Goal: Gated pipeline — tests block broken builds, versioned images enable rollback, RollingUpdate gives zero-downtime, Service exposes app. Jenkins automates GitHub webhook → Test → Build → Push → Deploy.

Directory Tree

```

cicd-pipeline/ (remote: cicd-python-app-k8s)
├── app/app.py           # Flask routes
├── app/requirements.txt # deps
├── app/test_app.py      # pytest gate
├── Dockerfile           # python:3.11-slim + gunicorn
├── k8s/deployment.yaml  # 3 replicas RollingUpdate sravani81299:1.0.1
├── k8s/service.yaml     # LoadBalancer 80->5000
├── jenkins/Jenkinsfile  # pipeline 5 stages
└── .gitignore, README.md, architecture.png/drawio, screenshots/

```

app/app.py

Flask app: / returns Hello from CI/CD Pipeline - sravani-devopsVersion: 1.0.1, /health jsonify status ok version 1.0.1, /api/info app sravani-devops env dev, run host 0.0.0.0 port 5000. Interview: '/health is for liveness/readiness probes; version bump tests rolling update.'

cicd-pipeline/app/app.py

```

from flask import Flask, jsonify

app = Flask(__name__)

@app.route("/")
def home():
    return "<h1>Hello from CI/CD Pipeline - sravani-devops</h1><p>Version: 1.0.1 - Rolling Update Demo</p>"

@app.route("/health")
def health():
    return jsonify(status="ok", version="1.0.1")

@app.route("/api/info")
def info():
    return jsonify(app="sravani-devops", env="dev")

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)

```

app/requirements.txt

flask==3.1.0, gunicorn==23.0.0, pytest==8.3.0. Pinned to avoid breaking TF. pip install --no-cache-dir.

cicd-pipeline\app\requirements.txt

```
flask==3.1.0
unicorn==23.0.0
pytest==8.3.0
```

app/test_app.py

Tests gate: test_health GET /health 200 status ok, test_home GET / 200 contains CI/CD Pipeline. Jenkins Test stage runs pip install + pytest; if fails, Build/Push/Deploy never run. Interview: 'Tests are the quality gate; I intentionally did not assert version so 1.0.0→1.0.1 doesn't break.'

cicd-pipeline\app\test_app.py

```
from app import app

def test_health():
    client = app.test_client()
    resp = client.get("/health")
    assert resp.status_code == 200
    data = resp.get_json()
    assert data["status"] == "ok"

def test_home():
    client = app.test_client()
    resp = client.get("/")
    assert resp.status_code == 200
    assert b"CI/CD Pipeline" in resp.data
```

Dockerfile

FROM python:3.11-slim, WORKDIR /app, COPY requirements.txt + pip install, COPY app/ ., EXPOSE 5000, CMD unicorn --bind 0.0.0.0:5000 app:app. Interview: 'Slim keeps image 217MB; unicorn for prod vs flask dev.'

cicd-pipeline\Dockerfile

```
FROM python:3.11-slim
WORKDIR /app
COPY app/requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app/ .
EXPOSE 5000
CMD ["unicorn", "--bind", "0.0.0.0:5000", "app:app"]
```

k8s/deployment.yaml

apps/v1 Deployment sravani-app 3 replicas strategy RollingUpdate maxSurge1 maxUnavailable1, selector app sravani-app, template containers app image sravani81299/sravani-app:1.0.1 port 5000 livenessProbe /health initialDelay 15 period 10 readinessProbe 5/5. Jenkins replaces image via sed + kubectl set image. Interview: 'RollingUpdate keeps 2/3 available, probes ensure traffic only to ready pods.'

cicd-pipeline\k8s\deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: sravani-app
  labels: { app: sravani-app }
spec:
  replicas: 3
  strategy:
    type: RollingUpdate
    rollingUpdate: { maxSurge: 1, maxUnavailable: 1 }
  selector: { matchLabels: { app: sravani-app } }
  template:
    metadata: { labels: { app: sravani-app } }
    spec:
      containers:
        - name: app
          image: sravani81299/sravani-app:1.0.0 # replaced by Jenkins
          ports: [{ containerPort: 5000 }]
          livenessProbe:
            httpGet: { path: /health, port: 5000 }
            initialDelaySeconds: 15
            periodSeconds: 10
          readinessProbe:
            httpGet: { path: /health, port: 5000 }
            initialDelaySeconds: 5
            periodSeconds: 5
```

k8s/service.yaml

Service sravani-app-svc selector app sravani-app ports 80→5000 TCP type LoadBalancer (ClusterIP+Ingress cheaper for EKS). kind → port-forward 8080:80. Interview: 'LoadBalancer gives AWS ELB; local kind uses port-forward.'

cicd-pipeline\k8s\service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: sravani-app-svc
spec:
  selector: { app: sravani-app }
  ports:
    - port: 80
      targetPort: 5000
      protocol: TCP
  type: LoadBalancer # use ClusterIP + Ingress for EKS cost saving
```

jenkins/Jenkinsfile

Pipeline: agent any, env DOCKERHUB_CREDS credentials('dockerhub') IMAGE sravani81299/sravani-app:\${BUILD_NUMBER} KUBECONFIG, stages Checkout scm, Test pip+pytest, Build docker build -t \$IMAGE, Push docker login + push \$IMAGE + tag latest, Deploy sed s/sravani81299.*/\$IMAGE/ k8s/deployment.yaml; kubectl apply -f k8s/; rollout status, post docker logout. Interview: 'BUILD_NUMBER versions images for rollback; latest for convenience; kubectl rollout status blocks until ready.'

cicd-pipeline\jenkins\Jenkinsfile

```
pipeline {
  agent any
  environment {
    DOCKERHUB_CREDS = credentials('dockerhub')
    IMAGE = "sravani81299/sravani-app:${env.BUILD_NUMBER}"
    KUBECONFIG = credentials('kubeconfig')
  }
  stages {
    stage('Checkout') { steps { checkout scm } }
    stage('Test') {
      steps {
        sh 'pip install -r app/requirements.txt && pytest app/test_app.py -v'
      }
    }
    stage('Build Docker') {
      steps { sh 'docker build -t $IMAGE .' }
    }
    stage('Push to DockerHub/ECR') {
      steps {
        sh 'echo $DOCKERHUB_CREDS_PSW | docker login -u $DOCKERHUB_CREDS_USR --password-stdin'
        sh 'docker push $IMAGE'
        sh 'docker tag $IMAGE sravani81299/sravani-app:latest && docker push sravani81299/sravani-app:latest'
      }
    }
    stage('Deploy to K8s') {
      steps {
        sh 'sed -i "s|sravani81299/sravani-app:.*|$IMAGE|g" k8s/deployment.yaml'
        sh 'kubectl apply -f k8s/deployment.yaml'
        sh 'kubectl apply -f k8s/service.yaml'
        sh 'kubectl rollout status deployment/sravani-app'
      }
    }
  }
  post {
    always { sh 'docker logout' }
  }
}
```

.gitignore

Full Python template __pycache__/, *.py[cod], .pytest_cache/, .venv/, venv/ + Node node_modules/ + Secrets .env, *.pem, .kube/, + OS .DS_Store. Never commit secrets.

cicd-pipeline\gitignore

```
# Python — official gitignore template
__pycache__/
*.py[cod]
*$py.class
*.so
```

```
.Python
build/
develop-eggs/
dist/
downloads/
eggs/
.eggs/
lib/
lib64/
parts/
sdist/
var/
wheels/
*.egg-info/
.installed.cfg
*.egg
MANIFEST
.pytest_cache/
.coverage
htmlcov/
.tox/
.nox/
.hypothesis/
.venv/
venv/
env/
ENV/
env.bak/
venv.bak/
pip-log.txt
pip-delete-this-directory.txt

# Node (if frontend added)
node_modules/
npm-debug.log*
... (29 more lines)
```

6. Commands Executed — Copy-Paste Log

All commands as run on 2026-08-29 (PowerShell + WSL). Replace < > with your values. **Order: Terraform → Ansible → CI/CD.**

Prerequisites

```
winget install --id Hashicorp.Terraform -e
winget install --id Docker.DockerDesktop -e
winget install --id Kubernetes.kind -e
winget install --id GitHub.cli -e
terraform version # 1.15.8
aws sts get-caller-identity # terraform_user 968153778824 us-east-1
docker --version # 29.7.2 desktop-linux
py -m pip install -r app/requirements.txt
```

Terraform Init & Fix Locks

```
cd terraform-aws-infra
aws s3api put-bucket-versioning --bucket sravani-terraform-demo-bucket1 --versioning-
configuration Status=Enabled
terraform fmt -recursive
terraform init -reconfigure
terraform validate
# If 412 lock: aws s3api list-object-versions --bucket sravani-terraform-demo-bucket1 --prefix
terraform-aws-infra/terraform.tfstate.tflock
# aws s3api delete-object --bucket ... --key ... --version-id <ID> (all versions)
# terraform force-unlock <ID> # e.g. 37fa9f3f...
terraform plan -out=tfplan
```

Terraform Apply (t3.micro fix)

```
# First plan 16 to add, apply 15 ok, EC2 failed t2.micro not free-tier -> change variables.tf:
42 t2.micro->t3.micro
terraform apply tfplan # or -auto-approve
terraform output # vpc-03b31ff28e41bc125, 3.231.160.92, sg-063451d20929be4f5
curl http://3.231.160.92/ # <h1>sravani-devops - dev - ip-10-0-1-213.ec2.internal</h1>
# For Ansible SSH
aws ec2 create-key-pair --key-name sravani-key --query KeyMaterial --output text > ~/.ssh/
sravani-key.pem
# set terraform.tfvars key_name="sravani-key" -> terraform apply -> new EC2 i-
00f507eb0af08bb0e 3.231.160.92
```


Ansible (WSL)

```
# WSL pip (Ubuntu 24.04 has no pip)
curl -sS https://bootstrap.pypa.io/get-pip.py -o /tmp/get-pip.py
python3 /tmp/get-pip.py --user --break-system-packages
~/local/bin/pip install --user --break-system-packages ansible
~/local/bin/ansible --version # 2.21.3
cp /mnt/c/Users/srava/.ssh/sravani-key.pem ~/.ssh/ && chmod 400 ~/.ssh/sravani-key.pem
# inventory already 3.231.160.92
ANSIBLE_ROLES_PATH=./roles ~/local/bin/ansible all -m ping -i inventory/hosts.ini # pong
ANSIBLE_ROLES_PATH=./roles ~/local/bin/ansible-playbook playbooks/site.yml -i inventory/hosts.
ini --check --diff
ANSIBLE_ROLES_PATH=./roles ~/local/bin/ansible-playbook playbooks/site.yml -i inventory/hosts.
ini # ok=12 changed=2
curl http://3.231.160.92/ # Configured by Ansible - web-1
curl http://3.231.160.92/health # ok
# firewall fix if Connection refused: ssh ec2-user@3.231.160.92 "sudo firewall-cmd --add-
service=http --permanent; sudo firewall-cmd --reload"
```

CI/CD Local

```
cd cicd-pipeline
py -m pip install -r app/requirements.txt
py -m pytest app/test_app.py -v # 2 passed
docker build -t sravani81299/sravani-app:1.0.0 -t sravani81299/sravani-app:latest .
docker run -d -p 5000:5000 sravani81299/sravani-app:1.0.0; curl http://localhost:5000/health #
ok
docker login -u sravani81299 # paste PAT
docker push sravani81299/sravani-app:1.0.0
# Bump app.py 1.0.0->1.0.1, pytest, docker build -t 1.0.1
kind create cluster --name sravani-kind --wait 3m # v1.37.0
kind load docker-image sravani81299/sravani-app:1.0.0 --name sravani-kind
kubectl apply -f k8s/deployment.yaml; kubectl apply -f k8s/service.yaml # sravani81299:1.0.0
kubectl rollout status deployment/sravani-app # 3/3 Running
kubectl port-forward svc/sravani-app-svc 8080:80 & curl http://localhost:8080/health # ok
kind load docker-image sravani81299/sravani-app:1.0.1 --name sravani-kind
kubectl set image deployment/sravani-app app=sravani81299/sravani-app:1.0.1
kubectl rollout status deployment/sravani-app # 1.0.1 rolling update
curl http://localhost:8080/ # Version: 1.0.1
docker tag sravani81299/sravani-app:1.0.1 sravani81299/sravani-app:latest; docker push
sravani81299/sravani-app:1.0.1; docker push latest
kind delete cluster --name sravani-kind
```

Git Push

```
cd terraform-aws-infra; git add .; git commit -m "feat: ..."; git branch -M main; git remote
add origin https://github.com/sravanigaddam-devops/terraform-aws-infra.git; git push -u origin
main
cd ../ansible-config-mgmt; git add .; git commit -m "feat: ..."; git branch -M main; git
remote add origin https://github.com/sravanigaddam-devops/ansible-config-mgmt.git; git push -u
origin main
cd ../cicd-pipeline; git add .; git commit -m "feat: ..."; git branch -M main; git remote add
origin https://github.com/sravanigaddam-devops/cicd-python-app-k8s.git; git push -u origin main
# If non-fast-forward: git push --force origin main (initial scaffold divergence)
```

Jenkins (Optional)

```
# Install Jenkins + plugins Docker, K8s, GitHub, Pipeline
# Manage Credentials: dockerhub (sravani81299 PAT) KUBECONFIG (kind config)
# New Pipeline → GitHub hook trigger → SCM https://github.com/sravanigaddam-devops/cicd-python-
app-k8s.git Script Path jenkins/Jenkinsfile
# GitHub → Settings → Webhooks → http://<jenkins>:8080/github-webhook/
# Push → webhook → Test→Build→Push→Deploy
```

Cleanup

```
cd terraform-aws-infra; terraform destroy -auto-approve # 16 destroyed
kind delete cluster --name sravani-kind
# Keep GitHub + Docker Hub + ~/.ssh/sravani-key.pem
```

7. Live Outputs & Proof

All outputs captured 2026-08-29 before *terraform destroy*. See *screenshots/*.txt* in each repo.

Project	Outputs
---------	---------

Terraform	ec2_instance_id=i-00f507eb0af08bb0e ec2_public_ip=3.231.160.92 vpc_id=vpc-03b31ff28e41bc125 public_subnet_ids subnet-0dd6f2f25c1f7f34c, 03888aed972cae226 terraform state list 16 resources (data aws_ami, eip, igw, nat, rt, subnets, vpc, sg, ec2)
Ansible	ansible ping → pong playbook → ok=12 changed=2 failed=0 curl http://3.231.160.92/ → Configured by Ansible - web-1 curl /health → ok firewalld http/https open
CI/CD	pytest 2 passed docker images sravani81299:1.0.1 217MB sha256:18d0... Docker Hub 1.0.0/1.0.1/latest pushed kind v1.37.0 3/3 Running port-forward localhost:8080 → Version:1.0.1, /health ok rollout history REVISION 2

Screenshots prove beats prose: each repo *screenshots/* contains *terraform-output.txt*, *kubectctl-get-pods.txt* 3/3 Running, *docker-images.txt*, *curl-k8s.txt*, etc. + browser screenshot *http://3.231.160.92* Not secure Configured by Ansible (http expected). Architecture.png in all 3.

8. Troubleshooting — 7 Real Errors Fixed

412 PreconditionFailed .tflock

Versioned bucket left delete markers → `aws s3api list-object-versions + delete-object --version-id (all) + terraform force-unlock <ID>`

t2.micro not free-tier

`describe-instance-types --filters free-tier-eligible` showed t3.micro → `variables.tf:42 t2.micro→t3.micro`

ansible.cfg ignored

`ANSIBLE_ROLES_PATH=./roles` + cd to repo root

handler 'Restart sshd' not found

Added `roles/hardening/handlers/main.yml`

copy: notify unsupported

Moved `notify`: from inside `copy`: to task level (2 spaces)

Connection refused 3.231.160.92:80

firewalld started but no http → `firewall-cmd --add-service=http --permanent --reload` + role fix

WSL pip No module named pip

`curl get-pip.py --user --break-system-packages → ~/.local/bin/pip install --break-system-packages ansible`

9. Interview Guide — 30s / 2 / 5 / 10 min + Q&A;

30s Elevator: “Terraform S3 remote VPC, Ansible hardening/nginx on 3.231.160.92, gated Docker Hub + kind 3-replica RollingUpdate 1.0.1 — all live-verified, repos pushed, infra destroyed.”

2-min: Order 2→3→1, mention S3 lock, t3.micro, firewalld, pytest gate, BUILD_NUMBER version, rollout.

5-min: Add file refs: `provider.tf:1 backend.tf:1 modules/vpc/main.tf:1 inventory/hosts.ini:1 roles/webserver/tasks/main.yml:7 Dockerfile:1 k8s/deployment.yaml:7 Jenkinsfile:10` + one STAR story (state lock).

10-min Deep Dive: Walk `provider→modules→outputs→inventory→playbooks→app.py→requirements→test→Dockerfile→deployment→service→Jenkinsfile`, show *architecture.png*, curl outputs, `s3api list-object-versions`, `kubectl rollout history`.

Q: Why S3 native lock vs DynamoDB? A: TF >=1.10 S3 use `_lockfile` uses `If-None-Match`, no extra table, but versioned bucket needs delete-marker cleanup. **Q: Why kind vs EKS?** A: kind saves cost; swap to EKS by changing `kubeconfig` + ECR

login.

STAR — State Lock: S: plan 412, T: unblock without -lock=false, A: list/delete all .tflock versions + force-unlock, R: plan succeeded, documented. **STAR — Free-tier:** S: t2.micro rejected, T: find eligible, A: describe-instance-types, R: t3.micro applied.

10. Repos, Cleanup & Next Steps

GitHub (pushed 2026-08-29):

- <https://github.com/sravanigaddam-devops/terraform-aws-infra/blob/main/INTERVIEW.md> (cf6a68b)
- <https://github.com/sravanigaddam-devops/ansible-config-mgmt/blob/main/INTERVIEW.md> (728d0b8)
- <https://github.com/sravanigaddam-devops/cicd-python-app-k8s/blob/main/INTERVIEW.md> (c189ebd)

Docker Hub: <https://hub.docker.com/r/sravani81299/sravani-app/tags> (1.0.0 sha256:77eb..., 1.0.1 sha256:18d0..., latest)

Local: architecture.png + generate_diagram.py:1 + DevOps_End_To_End_Documentation.pdf (this file) + screenshots/*.txt

Cleanup (done): terraform destroy -auto-approve # 16 destroyed, output empty; kind delete cluster --name sravani-kind; docker ps -a empty. Keep GitHub/Docker Hub/~/.ssh/sravani-key.pem. Recreate: terraform apply; kind create cluster; kubectl apply. Cost while live: NAT GW \$0.045/hr + t3.micro.

.gitignore: Python/Node template — no node_modules/, __pycache__/, .venv/, *.pyc, .pytest_cache/, .env, *.pem, .terraform/, *.tfstate* (keeps !terraform.tfvars.example).

Generated 2026-08-29 from live runs. All file contents and commands are exact as executed. Proof screenshots + architecture.png in each repo. Contact: sravani81299@gmail.com